

Dear Helsing Team,

I'm applying for the Reinforcement Learning Engineer role on the Centaur project. I want to be direct about where I stand: my thesis work is deep RL applied to knowledge graph repair, not flight control, and I haven't worked with PPO/SAC, continuous control, or real-time deployment on flight hardware. I'd rather say that upfront than have it surface later.

What I can point to is real, hands-on RL engineering. My Master's thesis at RPTU Kaiserslautern formulates an autonomous-decision problem as a Markov Decision Process and trains a Double DQN agent with a dueling architecture to solve it: I built the 6-action environment myself, engineered shaped rewards and action masking to keep the agent within valid decisions, and used prioritised experience replay to stabilise training. I validated the learned policy against two independent formal reasoners as ground truth rather than trusting a single offline metric, and ran a three-phase human-in-the-loop protocol to move the agent from expert-guided learning to fully autonomous operation - achieving 88-96% task success at 5.1 steps per episode on a 2,000+ node industrial graph.

At Fraunhofer IGCV, I owned a full ML lifecycle for a live manufacturing system - training, Docker deployment, and ongoing performance monitoring for a model at ~90% accuracy - so I understand what it takes for a model to keep working after the demo ends, under real operational conditions, not just in a notebook.

What's missing for me right now is the specific technique set (PPO/SAC, multi-agent RL, population-based training), systems-level Rust/C++, and any robotics or flight dynamics background. What I do have is the underlying discipline this role is built on: formal problem structuring, rigorous evaluation against ground truth, and taking an agent from supervised training to autonomous deployment.

I'd welcome the chance to talk about where that's useful to you.

Best regards,

Resham Joshi